

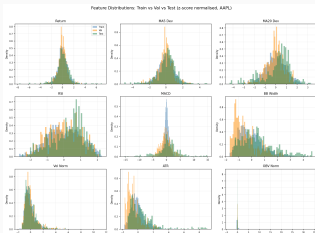
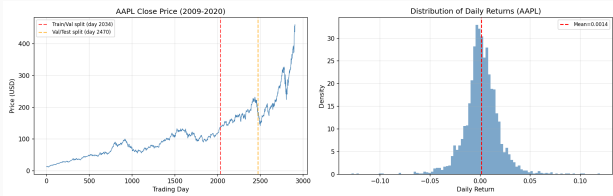
DQN & DDQN for Automated Stock Trading

Implementation, Iterative Improvements & Results

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Yufan: Data, Environment & Initial System



Train / val / test distributions overlap after train-only z-score normalisation, confirming no look-ahead bias.

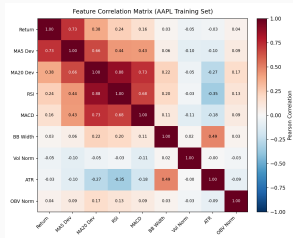
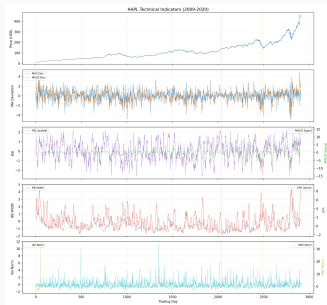
Reproducible market-data pipeline

- Daily OHLCV via Kaggle [alincijov/trading](#) ([kagglehub](#)).
- DJIA ticker subset, AAPL as case study (2009–2024).
- Chronological 70 / 15 / 15 train-val-test split.
- Raw returns preserved for reward & portfolio accounting.
- Z-score normalisation uses **training stats only**.

Why it matters

A leakage-safe pipeline is the prerequisite for honest RL evaluation: the test split must look like a true out-of-sample future to the agent. Same data, same split, same accounting for every model downstream.

Yufan — Technical Indicators



Nine engineered indicators

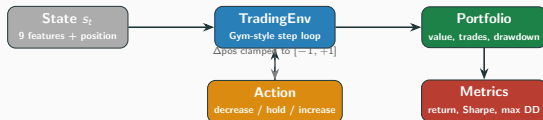
Group	Features
Trend	Return, MA5 dev., MA20 dev.
Momentum	RSI-14, MACD signal
Volatility	Bollinger width, ATR
Volume	Normalised volume, OBV

Design rationale

- Each family captures a distinct market regime.
- Correlation heatmap shows mostly low cross-correlation — features carry complementary signal.
- Strongest pair: BB width vs ATR (≈ 0.63), both encode volatility.

Goal: a clean 9-dimensional state vector that lets the agent learn from price, momentum, volatility and volume jointly.

Yufan — Trading Environment & Training



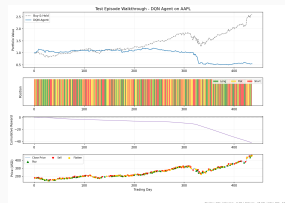
$$V_t = V_{t-1}(1 + \text{pos}_t \tilde{r}_t - c_t 1[\Delta \text{pos} \neq 0])$$
$$r_t = \log(V_t / V_{t-1}) + \lambda \min(0, \text{drawdown}_t)$$

Environment design

- Positions: short, flat, long; actions shift by one step.
- Transaction cost: 10 bps per position change.
- Drawdown penalty discourages fragile portfolios.

Training loop

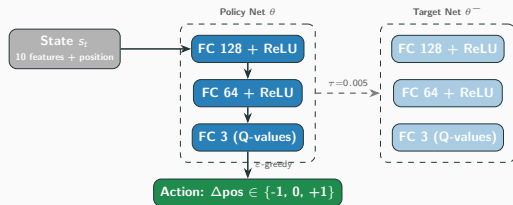
- Each episode replays the full $\sim 2,800$ -day train split.
- Greedy validation rollout every 10 episodes ($\varepsilon = 0$).
- Sharpe, max DD, trade count computed at episode end.



Walkthrough links actions to portfolio value, cumulative reward and trade timing.

Rutwik: DQN/DDQN Refinements & Results

DQN & DDQN — Architecture & Implementation



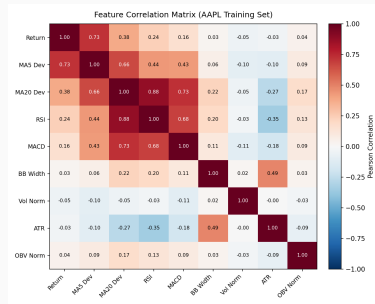
DDQN: select $a^* = \arg \max Q_{\theta^-}(s', a^*)$

DQN: Replay buffer breaks temporal correlations; target network provides stable TD targets. Counterfactual replay improves sample efficiency.

DDQN: Decouples action *selection* (policy net) from *evaluation* (target net) to reduce overestimation bias.

10 Technical Features (z-score normalised):

Feature	Signal Type
Return, Dev5, Dev20	Trend / mean-reversion
Mom20	Momentum
RSI, MACD	Oscillator / crossover
BB Width, ATR	Volatility
Vol Norm, OBV Norm	Volume flow



Volume/volatility indicators are near-uncorrelated with trend indicators, suggesting reduced redundancy and more diverse signals.

Iterative Improvements Across Versions

Component	V1 (Initial)	V2 (Core)	V4/HEAD
Actions	Set position	Delta-based	Delta-based
Reward	Raw PnL	Log-return	Log-return
ϵ	Linear, 1.0 $\rightarrow$.05	Exp decay	0.2 $\rightarrow$.02
Target upd.	Hard (N=100)	Soft $\tau=.005$	Soft
Loss	MSE	Huber	Huber
LR / Batch	1e-3 / 64	5e-4 / 128	1e-4 / 128
Buffer	10 k	50 k	50 k
Episodes	Full dataset	252d random	252d random
Warm start	—	—	Momentum
Replay	Standard	Standard	+Counterfact.

Green = change introduced in that version.

Three most impactful changes:

1. Reward: Raw PnL $\rightarrow$ Log-Return

$$r_t = \log\left(\frac{V_{t+1}}{V_t}\right) \quad \text{Scale-invariant, additive}$$

Raw PnL was dominated by large price moves; log-return treats gains/losses symmetrically

and is additive over time, improving numerical stability for sequential optimisation.

2. Supervised Warm Start from Momentum Teacher

Pre-train Q-net via cross-entropy on momentum signal (20d lookback). Builds (s, a) pairs for

every reachable position so policy learns entry & exit. Lower $\epsilon_0=0.2$ preserves pre-trained weights.

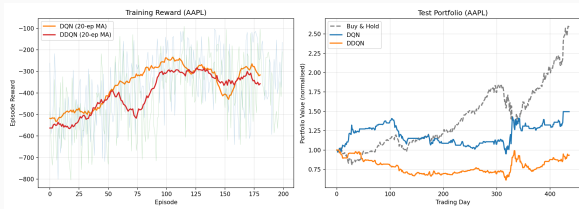
3. TDQN-Style Counterfactual Experience Replay

For each step, compute exact rewards for the 2 actions **not** taken. Pushes $3\times$ transitions per

step $\rightarrow$ significantly improves sample efficiency in practice. Possible in our deterministic

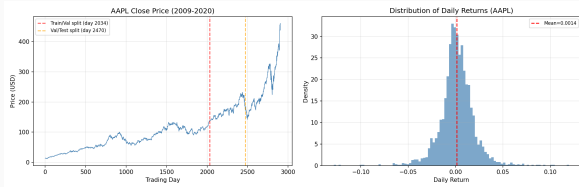
backtesting environment, where rewards depend only on price data.

Results & Analysis (AAPL, 2018–2020 Test Set)



Left: Training reward (20-ep MA). Agents generally trend upward. Right: Test portfolio value.

Modern RL agents (PPO, A2C, DDQN) consistently outpace Buy & Hold.



AAPL 2009–2020 with train/val/test splits. Test period is a strong bull market.

Test Performance:

Model	Return	Sharpe	Max DD
PPO	+184.3%	2.81	−16.7%
A2C	+182.9%	2.79	−17.1%
DDQN	+172.5%	2.73	−15.9%
DQN	+171.7%	2.73	−15.5%
Momentum	+167.0%	2.62	−17.5%
Buy & Hold	+159.3%	1.65	−31.4%
ARIMA	+121.1%	1.67	−18.8%
Random Forest	+81.1%	1.25	−26.8%

Key Takeaways:

- ✓ **RL outperforms baselines:** All RL agents beat Buy & Hold despite strong bull market constraints.
- ✓ **Actor-Critic superiority:** PPO and A2C yield top returns, better managing noisy financial sequences.
- ✓ **DDQN $\gtrsim$ DQN:** DDQN slightly outperforms DQN, consistent with reduced overestimation bias.
- ! RL models exhibit lower max drawdowns ($\sim 16\%$), roughly half that of B&H.

Fei Xin: Actor-Critic Methods
